

Chapter 3

Random Variables and Probability Distributions

3.1 Concept of a Random Variable

Statistics is concerned with making inferences about populations and population characteristics. Experiments are conducted with results that are subject to chance. The testing of a number of electronic components is an example of a **statistical experiment**, a term that is used to describe any process by which several chance observations are generated. It is often important to allocate a numerical description to the outcome. For example, the sample space giving a detailed description of each possible outcome when three electronic components are tested may be written

$$S = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\},$$

where N denotes nondefective and D denotes defective. One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be *assigned a numerical value* of 0, 1, 2, or 3. These values are, of course, random quantities *determined by the outcome of the experiment*. They may be viewed as values assumed by the *random variable* X , the number of defective items when three electronic components are tested.

Definition 3.1: A **random variable** is a function that associates a real number with each element in the sample space.

We shall use a capital letter, say X , to denote a random variable and its corresponding small letter, x in this case, for one of its values. In the electronic component testing illustration above, we notice that the random variable X assumes the value 2 for all elements in the subset

$$E = \{DDN, DND, NDD\}$$

of the sample space S . That is, each possible value of X represents an event that is a subset of the sample space for the given experiment.

Example 3.1: Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y , where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0

Example 3.2: A stockroom clerk returns three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Solution: If S , J , and B stand for Smith's, Jones's, and Brown's helmets, respectively, then the possible arrangements in which the helmets may be returned and the number of correct matches are

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
JBS	0
BSJ	0

In each of the two preceding examples, the sample space contains a finite number of elements. On the other hand, when a die is thrown until a 5 occurs, we obtain a sample space with an unending sequence of elements,

$$S = \{F, NF, NNF, NNNF, \dots\},$$

where F and N represent, respectively, the occurrence and nonoccurrence of a 5. But even in this experiment, the number of elements can be equated to the number of whole numbers so that there is a first element, a second element, a third element, and so on, and in this sense can be counted.

There are cases where the random variable is categorical in nature. Variables, often called *dummy* variables, are used. A good illustration is the case in which the random variable is binary in nature, as shown in the following example.

Example 3.3: Consider the simple condition in which components are arriving from the production line and they are stipulated to be defective or not defective. Define the random variable X by

$$X = \begin{cases} 1, & \text{if the component is defective,} \\ 0, & \text{if the component is not defective.} \end{cases}$$

Clearly the assignment of 1 or 0 is arbitrary though quite convenient. This will become clear in later chapters. The random variable for which 0 and 1 are chosen to describe the two possible values is called a **Bernoulli random variable**. ■

Further illustrations of random variables are revealed in the following examples.

Example 3.4: Statisticians use **sampling plans** to either accept or reject batches or lots of material. Suppose one of these sampling plans involves sampling independently 10 items from a lot of 100 items in which 12 are defective.

Let X be the random variable defined as the number of items found defective in the sample of 10. In this case, the random variable takes on the values 0, 1, 2, ..., 9, 10. ■

Example 3.5: Suppose a sampling plan involves sampling items from a process until a defective is observed. The evaluation of the process will depend on how many consecutive items are observed. In that regard, let X be a random variable defined by the number of items observed before a defective is found. With N a nondefective and D a defective, sample spaces are $S = \{D\}$ given $X = 1$, $S = \{ND\}$ given $X = 2$, $S = \{NND\}$ given $X = 3$, and so on. ■

Example 3.6: Interest centers around the proportion of people who respond to a certain mail order solicitation. Let X be that proportion. X is a random variable that takes on all values x for which $0 \leq x \leq 1$. ■

Example 3.7: Let X be the random variable defined by the waiting time, in hours, between successive speeders spotted by a radar unit. The random variable X takes on all values x for which $x \geq 0$. ■

Definition 3.2: If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**.

The outcomes of some statistical experiments may be neither finite nor countable. Such is the case, for example, when one conducts an investigation measuring the distances that a certain make of automobile will travel over a prescribed test course on 5 liters of gasoline. Assuming distance to be a variable measured to any degree of accuracy, then clearly we have an infinite number of possible distances in the sample space that cannot be equated to the number of whole numbers. Or, if one were to record the length of time for a chemical reaction to take place, once again the possible time intervals making up our sample space would be infinite in number and uncountable. We see now that all sample spaces need not be discrete.

Definition 3.3: If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

A random variable is called a **discrete random variable** if its set of possible outcomes is countable. The random variables in Examples 3.1 to 3.5 are discrete random variables. But a random variable whose set of possible values is an entire interval of numbers is not discrete. When a random variable can take on values

on a continuous scale, it is called a **continuous random variable**. Often the possible values of a continuous random variable are precisely the same values that are contained in the continuous sample space. Obviously, the random variables described in Examples 3.6 and 3.7 are continuous random variables.

In most practical problems, continuous random variables represent *measured* data, such as all possible heights, weights, temperatures, distance, or life periods, whereas discrete random variables represent *count* data, such as the number of defectives in a sample of k items or the number of highway fatalities per year in a given state. Note that the random variables Y and M of Examples 3.1 and 3.2 both represent count data, Y the number of red balls and M the number of correct hat matches.

3.2 Discrete Probability Distributions

A discrete random variable assumes each of its values with a certain probability. In the case of tossing a coin three times, the variable X , representing the number of heads, assumes the value 2 with probability $3/8$, since 3 of the 8 equally likely sample points result in two heads and one tail. If one assumes equal weights for the simple events in Example 3.2, the probability that no employee gets back the right helmet, that is, the probability that M assumes the value 0, is $1/3$. The possible values m of M and their probabilities are

m	0	1	3
$P(M = m)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

Note that the values of m exhaust all possible cases and hence the probabilities add to 1.

Frequently, it is convenient to represent all the probabilities of a random variable X by a formula. Such a formula would necessarily be a function of the numerical values x that we shall denote by $f(x)$, $g(x)$, $r(x)$, and so forth. Therefore, we write $f(x) = P(X = x)$; that is, $f(3) = P(X = 3)$. The set of ordered pairs $(x, f(x))$ is called the **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X .

Definition 3.4: The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,
3. $P(X = x) = f(x)$.

Example 3.8: A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Solution: Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and

2. Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}, \quad f(1) = P(X = 1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190},$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}.$$

Thus, the probability distribution of X is

x	0	1	2
$f(x)$	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

■

Example 3.9: If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.

Solution: Since the probability of selling an automobile with side airbags is 0.5, the $2^4 = 16$ points in the sample space are equally likely to occur. Therefore, the denominator for all probabilities, and also for our function, is 16. To obtain the number of ways of selling 3 cars with side airbags, we need to consider the number of ways of partitioning 4 outcomes into two cells, with 3 cars with side airbags assigned to one cell and the model without side airbags assigned to the other. This can be done in $\binom{4}{3} = 4$ ways. In general, the event of selling x models with side airbags and $4 - x$ models without side airbags can occur in $\binom{4}{x}$ ways, where x can be 0, 1, 2, 3, or 4. Thus, the probability distribution $f(x) = P(X = x)$ is

$$f(x) = \frac{1}{16} \binom{4}{x}, \quad \text{for } x = 0, 1, 2, 3, 4.$$

■

There are many problems where we may wish to compute the probability that the observed value of a random variable X will be less than or equal to some real number x . Writing $F(x) = P(X \leq x)$ for every real number x , we define $F(x)$ to be the **cumulative distribution function** of the random variable X .

Definition 3.5: The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

For the random variable M , the number of correct matches in Example 3.2, we have

$$F(2) = P(M \leq 2) = f(0) + f(1) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

The cumulative distribution function of M is

$$F(m) = \begin{cases} 0, & \text{for } m < 0, \\ \frac{1}{3}, & \text{for } 0 \leq m < 1, \\ \frac{5}{6}, & \text{for } 1 \leq m < 3, \\ 1, & \text{for } m \geq 3. \end{cases}$$

One should pay particular notice to the fact that the cumulative distribution function is a monotone nondecreasing function defined not only for the values assumed by the given random variable but for all real numbers.

Example 3.10: Find the cumulative distribution function of the random variable X in Example 3.9. Using $F(x)$, verify that $f(2) = 3/8$.

Solution: Direct calculations of the probability distribution of Example 3.9 give $f(0) = 1/16$, $f(1) = 1/4$, $f(2) = 3/8$, $f(3) = 1/4$, and $f(4) = 1/16$. Therefore,

$$\begin{aligned} F(0) &= f(0) = \frac{1}{16}, \\ F(1) &= f(0) + f(1) = \frac{5}{16}, \\ F(2) &= f(0) + f(1) + f(2) = \frac{11}{16}, \\ F(3) &= f(0) + f(1) + f(2) + f(3) = \frac{15}{16}, \\ F(4) &= f(0) + f(1) + f(2) + f(3) + f(4) = 1. \end{aligned}$$

Hence,

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{1}{16}, & \text{for } 0 \leq x < 1, \\ \frac{5}{16}, & \text{for } 1 \leq x < 2, \\ \frac{11}{16}, & \text{for } 2 \leq x < 3, \\ \frac{15}{16}, & \text{for } 3 \leq x < 4, \\ 1 & \text{for } x \geq 4. \end{cases}$$

Now

$$f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{3}{8}.$$

It is often helpful to look at a probability distribution in graphic form. One might plot the points $(x, f(x))$ of Example 3.9 to obtain Figure 3.1. By joining the points to the x axis either with a dashed or with a solid line, we obtain a probability mass function plot. Figure 3.1 makes it easy to see what values of X are most likely to occur, and it also indicates a perfectly symmetric situation in this case.

Instead of plotting the points $(x, f(x))$, we more frequently construct rectangles, as in Figure 3.2. Here the rectangles are constructed so that their bases of equal width are centered at each value x and their heights are equal to the corresponding probabilities given by $f(x)$. The bases are constructed so as to leave no space between the rectangles. Figure 3.2 is called a **probability histogram**.

Since each base in Figure 3.2 has unit width, $P(X = x)$ is equal to the area of the rectangle centered at x . Even if the bases were not of unit width, we could adjust the heights of the rectangles to give areas that would still equal the probabilities of X assuming any of its values x . This concept of using areas to represent

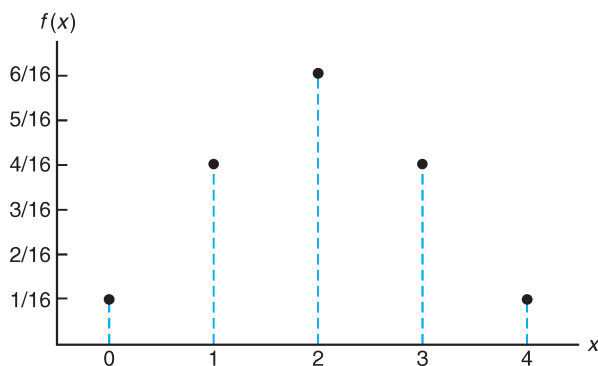


Figure 3.1: Probability mass function plot.

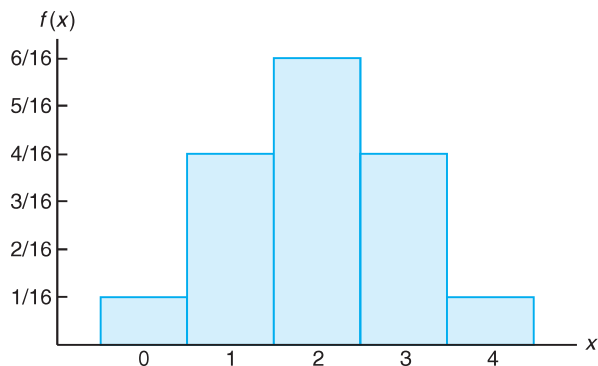


Figure 3.2: Probability histogram.

probabilities is necessary for our consideration of the probability distribution of a continuous random variable.

The graph of the cumulative distribution function of Example 3.9, which appears as a step function in Figure 3.3, is obtained by plotting the points $(x, F(x))$.

Certain probability distributions are applicable to more than one physical situation. The probability distribution of Example 3.9, for example, also applies to the random variable Y , where Y is the number of heads when a coin is tossed 4 times, or to the random variable W , where W is the number of red cards that occur when 4 cards are drawn at random from a deck in succession with each card replaced and the deck shuffled before the next drawing. Special discrete distributions that can be applied to many different experimental situations will be considered in Chapter 5.

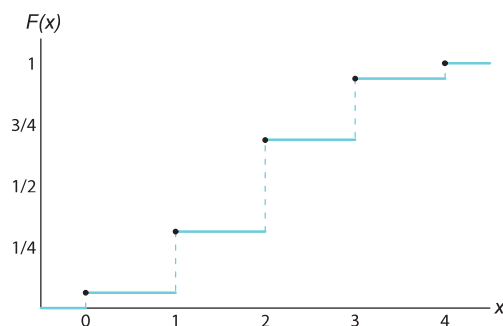


Figure 3.3: Discrete cumulative distribution function.

3.3 Continuous Probability Distributions

A continuous random variable has a probability of 0 of assuming *exactly* any of its values. Consequently, its probability distribution cannot be given in tabular form.

At first this may seem startling, but it becomes more plausible when we consider a particular example. Let us discuss a random variable whose values are the heights of all people over 21 years of age. Between any two values, say 163.5 and 164.5 centimeters, or even 163.99 and 164.01 centimeters, there are an infinite number of heights, one of which is 164 centimeters. The probability of selecting a person at random who is exactly 164 centimeters tall and not one of the infinitely large set of heights so close to 164 centimeters that you cannot humanly measure the difference is remote, and thus we assign a probability of 0 to the event. This is not the case, however, if we talk about the probability of selecting a person who is at least 163 centimeters but not more than 165 centimeters tall. Now we are dealing with an interval rather than a point value of our random variable.

We shall concern ourselves with computing probabilities for various intervals of continuous random variables such as $P(a < X < b)$, $P(W \geq c)$, and so forth. Note that when X is continuous,

$$P(a < X \leq b) = P(a < X < b) + P(X = b) = P(a < X < b).$$

That is, it does not matter whether we include an endpoint of the interval or not. This is not true, though, when X is discrete.

Although the probability distribution of a continuous random variable cannot be presented in tabular form, it can be stated as a formula. Such a formula would necessarily be a function of the numerical values of the continuous random variable X and as such will be represented by the functional notation $f(x)$. In dealing with continuous variables, $f(x)$ is usually called the **probability density function**, or simply the **density function**, of X . Since X is defined over a continuous sample space, it is possible for $f(x)$ to have a finite number of discontinuities. However, most density functions that have practical applications in the analysis of statistical data are continuous and their graphs may take any of several forms, some of which are shown in Figure 3.4. Because areas will be used to represent probabilities and probabilities are positive numerical values, the density function must lie entirely above the x axis.

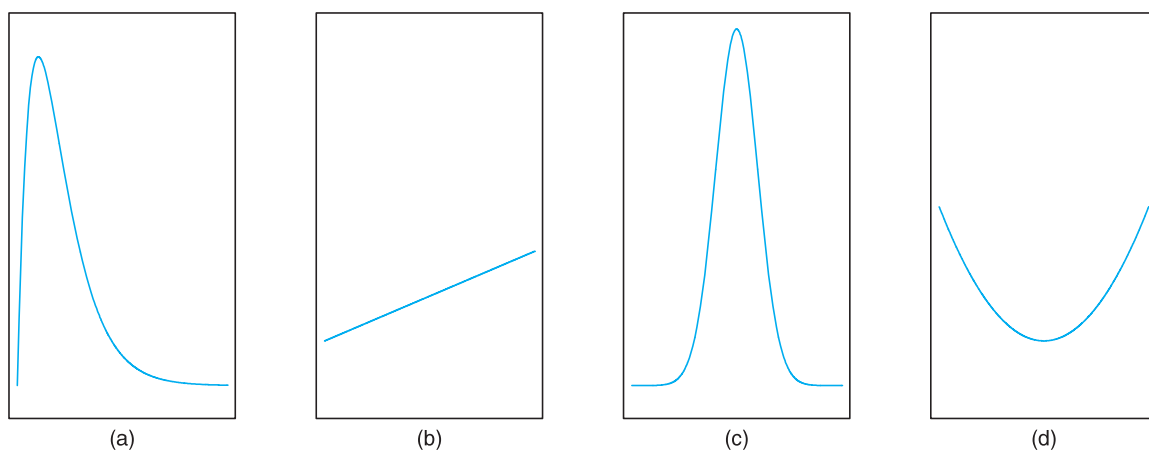


Figure 3.4: Typical density functions.

A probability density function is constructed so that the area under its curve

bounded by the x axis is equal to 1 when computed over the range of X for which $f(x)$ is defined. Should this range of X be a finite interval, it is always possible to extend the interval to include the entire set of real numbers by defining $f(x)$ to be zero at all points in the extended portions of the interval. In Figure 3.5, the probability that X assumes a value between a and b is equal to the shaded area under the density function between the ordinates at $x = a$ and $x = b$, and from integral calculus is given by

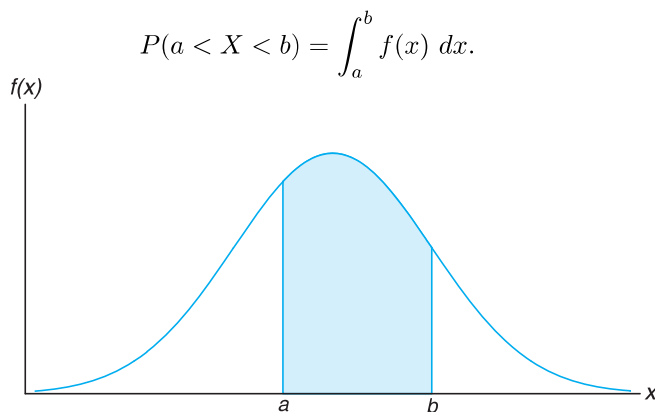


Figure 3.5: $P(a < X < b)$.

Definition 3.6: The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X , defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x) \, dx = 1$.
3. $P(a < X < b) = \int_a^b f(x) \, dx$.

Example 3.11: Suppose that the error in the reaction temperature, in $^{\circ}\text{C}$, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$.

Solution: We use Definition 3.6.

- (a) Obviously, $f(x) \geq 0$. To verify condition 2 in Definition 3.6, we have

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-1}^2 \frac{x^2}{3} \, dx = \frac{x^3}{9} \Big|_{-1}^2 = \frac{8}{9} + \frac{1}{9} = 1.$$

(b) Using formula 3 in Definition 3.6, we obtain

$$P(0 < X \leq 1) = \int_0^1 \frac{x^2}{3} dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}.$$



Definition 3.7:

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad \text{for } -\infty < x < \infty.$$

As an immediate consequence of Definition 3.7, one can write the two results

$$P(a < X < b) = F(b) - F(a) \text{ and } f(x) = \frac{dF(x)}{dx},$$

if the derivative exists.

Example 3.12:

For the density function of Example 3.11, find $F(x)$, and use it to evaluate $P(0 < X \leq 1)$.

Solution: For $-1 < x < 2$,

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-1}^x \frac{t^2}{3} dt = \frac{t^3}{9} \Big|_{-1}^x = \frac{x^3 + 1}{9}.$$

Therefore,

$$F(x) = \begin{cases} 0, & x < -1, \\ \frac{x^3 + 1}{9}, & -1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

The cumulative distribution function $F(x)$ is expressed in Figure 3.6. Now

$$P(0 < X \leq 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9},$$

which agrees with the result obtained by using the density function in Example 3.11. 

Example 3.13:

The Department of Energy (DOE) puts projects out on bid and generally estimates what a reasonable bid should be. Call the estimate b . The DOE has determined that the density function of the winning (low) bid is

$$f(y) = \begin{cases} \frac{5}{8b}, & \frac{2}{5}b \leq y \leq 2b, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $F(y)$ and use it to determine the probability that the winning bid is less than the DOE's preliminary estimate b .

Solution: For $2b/5 \leq y \leq 2b$,

$$F(y) = \int_{2b/5}^y \frac{5}{8b} dy = \frac{5t}{8b} \Big|_{2b/5}^y = \frac{5y}{8b} - \frac{1}{4}.$$

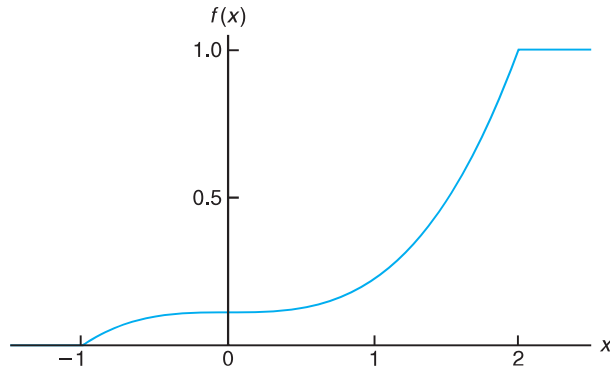


Figure 3.6: Continuous cumulative distribution function.

Thus,

$$F(y) = \begin{cases} 0, & y < \frac{2}{5}b, \\ \frac{5y}{8b} - \frac{1}{4}, & \frac{2}{5}b \leq y < 2b, \\ 1, & y \geq 2b. \end{cases}$$

To determine the probability that the winning bid is less than the preliminary bid estimate b , we have

$$P(Y \leq b) = F(b) = \frac{5}{8} - \frac{1}{4} = \frac{3}{8}.$$

■

Exercises

3.1 Classify the following random variables as discrete or continuous:

X : the number of automobile accidents per year in Virginia.

Y : the length of time to play 18 holes of golf.

M : the amount of milk produced yearly by a particular cow.

N : the number of eggs laid each month by a hen.

P : the number of building permits issued each month in a certain city.

Q : the weight of grain produced per acre.

3.2 An overseas shipment of 5 foreign automobiles contains 2 that have slight paint blemishes. If an agency receives 3 of these automobiles at random, list the elements of the sample space S , using the letters B and N for blemished and nonblemished, respectively;

then to each sample point assign a value x of the random variable X representing the number of automobiles with paint blemishes purchased by the agency.

3.3 Let W be a random variable giving the number of heads minus the number of tails in three tosses of a coin. List the elements of the sample space S for the three tosses of the coin and to each sample point assign a value w of W .

3.4 A coin is flipped until 3 heads in succession occur. List only those elements of the sample space that require 6 or less tosses. Is this a discrete sample space? Explain.

3.5 Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

(a) $f(x) = c(x^2 + 4)$, for $x = 0, 1, 2, 3$;

(b) $f(x) = c \binom{2}{x} \binom{3}{3-x}$, for $x = 0, 1, 2$.

3.6 The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

3.7 The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

- (a) less than 120 hours;
- (b) between 50 and 100 hours.

3.8 Find the probability distribution of the random variable W in Exercise 3.3, assuming that the coin is biased so that a head is twice as likely to occur as a tail.

3.9 The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the density function

$$f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Show that $P(0 < X < 1) = 1$.
- (b) Find the probability that more than $1/4$ but fewer than $1/2$ of the people contacted will respond to this type of solicitation.

3.10 Find a formula for the probability distribution of the random variable X representing the outcome when a single die is rolled once.

3.11 A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If x is the number of defective sets purchased by the hotel, find the probability distribution of X . Express the results graphically as a probability histogram.

3.12 An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

find

- (a) $P(T = 5)$;
- (b) $P(T > 3)$;
- (c) $P(1.4 < T < 6)$;
- (d) $P(T \leq 5 \mid T \geq 2)$.

3.13 The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct the cumulative distribution function of X .

3.14 The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X ;
- (b) using the probability density function of X .

3.15 Find the cumulative distribution function of the random variable X representing the number of defectives in Exercise 3.11. Then using $F(x)$, find

- (a) $P(X = 1)$;
- (b) $P(0 < X \leq 2)$.

3.16 Construct a graph of the cumulative distribution function of Exercise 3.15.

3.17 A continuous random variable X that can assume values between $x = 1$ and $x = 3$ has a density function given by $f(x) = 1/2$.

- (a) Show that the area under the curve is equal to 1.
- (b) Find $P(2 < X < 2.5)$.
- (c) Find $P(X \leq 1.6)$.

3.18 A continuous random variable X that can assume values between $x = 2$ and $x = 5$ has a density function given by $f(x) = 2(1 + x)/27$. Find

- (a) $P(X < 4)$;
- (b) $P(3 \leq X < 4)$.

3.19 For the density function of Exercise 3.17, find $F(x)$. Use it to evaluate $P(2 < X < 2.5)$.

3.20 For the density function of Exercise 3.18, find $F(x)$, and use it to evaluate $P(3 \leq X < 4)$.

3.21 Consider the density function

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Evaluate k .
- (b) Find $F(x)$ and use it to evaluate

$$P(0.3 < X < 0.6).$$

3.22 Three cards are drawn in succession from a deck without replacement. Find the probability distribution for the number of spades.

3.23 Find the cumulative distribution function of the random variable W in Exercise 3.8. Using $F(w)$, find

- (a) $P(W > 0)$;
- (b) $P(-1 \leq W < 3)$.

3.24 Find the probability distribution for the number of jazz CDs when 4 CDs are selected at random from a collection consisting of 5 jazz CDs, 2 classical CDs, and 3 rock CDs. Express your results by means of a formula.

3.25 From a box containing 4 dimes and 2 nickels, 3 coins are selected at random without replacement. Find the probability distribution for the total T of the 3 coins. Express the probability distribution graphically as a probability histogram.

3.26 From a box containing 4 black balls and 2 green balls, 3 balls are drawn in succession, each ball being replaced in the box before the next draw is made. Find the probability distribution for the number of green balls.

3.27 The time to failure in hours of an important piece of electronic equipment used in a manufactured DVD player has the density function

$$f(x) = \begin{cases} \frac{1}{2000} \exp(-x/2000), & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- (a) Find $F(x)$.
- (b) Determine the probability that the component (and thus the DVD player) lasts more than 1000 hours before the component needs to be replaced.
- (c) Determine the probability that the component fails before 2000 hours.

3.28 A cereal manufacturer is aware that the weight of the product in the box varies slightly from box to box. In fact, considerable historical data have allowed the determination of the density function that describes the probability structure for the weight (in ounces). Letting X be the random variable weight, in ounces, the density function can be described as

$$f(x) = \begin{cases} \frac{2}{5}, & 23.75 \leq x \leq 26.25, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that this is a valid density function.
- (b) Determine the probability that the weight is smaller than 24 ounces.
- (c) The company desires that the weight exceeding 26 ounces be an extremely rare occurrence. What is the probability that this rare occurrence does actually occur?

3.29 An important factor in solid missile fuel is the particle size distribution. Significant problems occur if the particle sizes are too large. From production data in the past, it has been determined that the particle size (in micrometers) distribution is characterized by

$$f(x) = \begin{cases} 3x^{-4}, & x > 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that this is a valid density function.
- (b) Evaluate $F(x)$.
- (c) What is the probability that a random particle from the manufactured fuel exceeds 4 micrometers?

3.30 Measurements of scientific systems are always subject to variation, some more than others. There are many structures for measurement error, and statisticians spend a great deal of time modeling these errors. Suppose the measurement error X of a certain physical quantity is decided by the density function

$$f(x) = \begin{cases} k(3 - x^2), & -1 \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Determine k that renders $f(x)$ a valid density function.
- (b) Find the probability that a random error in measurement is less than $1/2$.
- (c) For this particular measurement, it is undesirable if the *magnitude* of the error (i.e., $|x|$) exceeds 0.8. What is the probability that this occurs?

3.31 Based on extensive testing, it is determined by the manufacturer of a washing machine that the time Y (in years) before a major repair is required is characterized by the probability density function

$$f(y) = \begin{cases} \frac{1}{4}e^{-y/4}, & y \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

- Critics would certainly consider the product a bargain if it is unlikely to require a major repair before the sixth year. Comment on this by determining $P(Y > 6)$.
- What is the probability that a major repair occurs in the first year?

3.32 The proportion of the budget for a certain type of industrial company that is allotted to environmental and pollution control is coming under scrutiny. A data collection project determines that the distribution of these proportions is given by

$$f(y) = \begin{cases} 5(1-y)^4, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- Verify that the above is a valid density function.
- What is the probability that a company chosen at random spends less than 10% of its budget on environmental and pollution controls?
- What is the probability that a company selected at random spends more than 50% of its budget on environmental and pollution controls?

3.33 Suppose a certain type of small data processing firm is so specialized that some have difficulty making a profit in their first year of operation. The probability density function that characterizes the proportion Y that make a profit is given by

$$f(y) = \begin{cases} ky^4(1-y)^3, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- What is the value of k that renders the above a valid density function?
- Find the probability that at most 50% of the firms make a profit in the first year.
- Find the probability that at least 80% of the firms make a profit in the first year.

3.34 Magnetron tubes are produced on an automated assembly line. A sampling plan is used periodically to assess quality of the lengths of the tubes. This measurement is subject to uncertainty. It is thought that the probability that a random tube meets length specification is 0.99. A sampling plan is used in which the lengths of 5 random tubes are measured.

- Show that the probability function of Y , the number out of 5 that meet length specification, is given by the following discrete probability function:

$$f(y) = \frac{5!}{y!(5-y)!} (0.99)^y (0.01)^{5-y},$$

for $y = 0, 1, 2, 3, 4, 5$.

- Suppose random selections are made off the line and 3 are outside specifications. Use $f(y)$ above either to support or to refute the conjecture that the probability is 0.99 that a single tube meets specifications.

3.35 Suppose it is known from large amounts of historical data that X , the number of cars that arrive at a specific intersection during a 20-second time period, is characterized by the following discrete probability function:

$$f(x) = e^{-6} \frac{6^x}{x!}, \text{ for } x = 0, 1, 2, \dots$$

- Find the probability that in a specific 20-second time period, more than 8 cars arrive at the intersection.
- Find the probability that only 2 cars arrive.

3.36 On a laboratory assignment, if the equipment is working, the density function of the observed outcome, X , is

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Calculate $P(X \leq 1/3)$.
- What is the probability that X will exceed 0.5?
- Given that $X \geq 0.5$, what is the probability that X will be less than 0.75?

3.4 Joint Probability Distributions

Our study of random variables and their probability distributions in the preceding sections is restricted to one-dimensional sample spaces, in that we recorded outcomes of an experiment as values assumed by a single random variable. There will be situations, however, where we may find it desirable to record the simulta-

neous outcomes of several random variables. For example, we might measure the amount of precipitate P and volume V of gas released from a controlled chemical experiment, giving rise to a two-dimensional sample space consisting of the outcomes (p, v) , or we might be interested in the hardness H and tensile strength T of cold-drawn copper, resulting in the outcomes (h, t) . In a study to determine the likelihood of success in college based on high school data, we might use a three-dimensional sample space and record for each individual his or her aptitude test score, high school class rank, and grade-point average at the end of freshman year in college.

If X and Y are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by a function with values $f(x, y)$ for any pair of values (x, y) within the range of the random variables X and Y . It is customary to refer to this function as the **joint probability distribution** of X and Y .

Hence, in the discrete case,

$$f(x, y) = P(X = x, Y = y);$$

that is, the values $f(x, y)$ give the probability that outcomes x and y occur at the same time. For example, if an 18-wheeler is to have its tires serviced and X represents the number of miles these tires have been driven and Y represents the number of tires that need to be replaced, then $f(30000, 5)$ is the probability that the tires are used over 30,000 miles and the truck needs 5 new tires.

Definition 3.8:

The function $f(x, y)$ is a **joint probability distribution** or **probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y) ,
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

For any region A in the xy plane, $P[(X, Y) \in A] = \sum_A \sum f(x, y)$.

Example 3.14:

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

- (a) the joint probability function $f(x, y)$,
- (b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) | x + y \leq 1\}$.

Solution: The possible pairs of values (x, y) are $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, $(0, 2)$, and $(2, 0)$.

- (a) Now, $f(0, 1)$, for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$. The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1} \binom{3}{1} = 6$. Hence, $f(0, 1) = 6/28 = 3/14$. Similar calculations yield the probabilities for the other cases, which are presented in Table 3.1. Note that the probabilities sum to 1. In Chapter

5, it will become clear that the joint probability distribution of Table 3.1 can be represented by the formula

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}},$$

for $x = 0, 1, 2$; $y = 0, 1, 2$; and $0 \leq x + y \leq 2$.

(b) The probability that (X, Y) fall in the region A is

$$\begin{aligned} P[(X, Y) \in A] &= P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) \\ &= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14}. \end{aligned}$$

Table 3.1: Joint Probability Distribution for Example 3.14

$f(x, y)$		x			Row
		0	1	2	Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

When X and Y are continuous random variables, the **joint density function** $f(x, y)$ is a surface lying above the xy plane, and $P[(X, Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.

Definition 3.9:

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y) ,
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$, for any region A in the xy plane.

Example 3.15:

A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify condition 2 of Definition 3.9.
- (b) Find $P[(X, Y) \in A]$, where $A = \{(x, y) \mid 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$.

Solution: (a) The integration of $f(x, y)$ over the whole region is

$$\begin{aligned}\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy &= \int_0^1 \int_0^1 \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_0^1 \left(\frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{2}{5} + \frac{6y}{5} \right) dy = \left(\frac{2y}{5} + \frac{3y^2}{5} \right) \Big|_0^1 = \frac{2}{5} + \frac{3}{5} = 1.\end{aligned}$$

(b) To calculate the probability, we use

$$\begin{aligned}P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right) \\ &= \int_{1/4}^{1/2} \int_0^{1/2} \frac{2}{5}(2x + 3y) \, dx \, dy \\ &= \int_{1/4}^{1/2} \left(\frac{2x^2}{5} + \frac{6xy}{5} \right) \Big|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2} \left(\frac{1}{10} + \frac{3y}{5} \right) dy \\ &= \left(\frac{y}{10} + \frac{3y^2}{10} \right) \Big|_{1/4}^{1/2} \\ &= \frac{1}{10} \left[\left(\frac{1}{2} + \frac{3}{4} \right) - \left(\frac{1}{4} + \frac{3}{16} \right) \right] = \frac{13}{160}.\end{aligned}$$

Given the joint probability distribution $f(x, y)$ of the discrete random variables X and Y , the probability distribution $g(x)$ of X alone is obtained by summing $f(x, y)$ over the values of Y . Similarly, the probability distribution $h(y)$ of Y alone is obtained by summing $f(x, y)$ over the values of X . We define $g(x)$ and $h(y)$ to be the **marginal distributions** of X and Y , respectively. When X and Y are continuous random variables, summations are replaced by integrals. We can now make the following general definition.

Definition 3.10: The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.

The term *marginal* is used here because, in the discrete case, the values of $g(x)$ and $h(y)$ are just the marginal totals of the respective columns and rows when the values of $f(x, y)$ are displayed in a rectangular table.

Example 3.16: Show that the column and row totals of Table 3.1 give the marginal distribution of X alone and of Y alone.

Solution: For the random variable X , we see that

$$g(0) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show that the values of $h(y)$ are given by the row totals. In tabular form, these marginal distributions may be written as follows:

x	0	1	2
$g(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

y	0	1	2
$h(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

Example 3.17: Find $g(x)$ and $h(y)$ for the joint density function of Example 3.15.

Solution: By definition,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5}(2x + 3y) dy = \left(\frac{4xy}{5} + \frac{6y^2}{10} \right) \Big|_{y=0}^{y=1} = \frac{4x + 3}{5},$$

for $0 \leq x \leq 1$, and $g(x) = 0$ elsewhere. Similarly,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5}(2x + 3y) dx = \frac{2(1 + 3y)}{5},$$

for $0 \leq y \leq 1$, and $h(y) = 0$ elsewhere.

The fact that the marginal distributions $g(x)$ and $h(y)$ are indeed the probability distributions of the individual variables X and Y alone can be verified by showing that the conditions of Definition 3.4 or Definition 3.6 are satisfied. For example, in the continuous case

$$\int_{-\infty}^{\infty} g(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1,$$

and

$$\begin{aligned} P(a < X < b) &= P(a < X < b, -\infty < Y < \infty) \\ &= \int_a^b \int_{-\infty}^{\infty} f(x, y) dy dx = \int_a^b g(x) dx. \end{aligned}$$

In Section 3.1, we stated that the value x of the random variable X represents an event that is a subset of the sample space. If we use the definition of conditional probability as stated in Chapter 2,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) > 0,$$

where A and B are now the events defined by $X = x$ and $Y = y$, respectively, then

$$P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0,$$

where X and Y are discrete random variables.

It is not difficult to show that the function $f(x, y)/g(x)$, which is strictly a function of y with x fixed, satisfies all the conditions of a probability distribution. This is also true when $f(x, y)$ and $g(x)$ are the joint density and marginal distribution, respectively, of continuous random variables. As a result, it is extremely important that we make use of the special type of distribution of the form $f(x, y)/g(x)$ in order to be able to effectively compute conditional probabilities. This type of distribution is called a **conditional probability distribution**; the formal definition follows.

Definition 3.11: Let X and Y be two random variables, discrete or continuous. The **conditional distribution** of the random variable Y given that $X = x$ is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

If we wish to find the probability that the discrete random variable X falls between a and b when it is known that the discrete variable $Y = y$, we evaluate

$$P(a < X < b | Y = y) = \sum_{a < x < b} f(x|y),$$

where the summation extends over all values of X between a and b . When X and Y are continuous, we evaluate

$$P(a < X < b | Y = y) = \int_a^b f(x|y) dx.$$

Example 3.18: Referring to Example 3.14, find the conditional distribution of X , given that $Y = 1$, and use it to determine $P(X = 0 | Y = 1)$.

Solution: We need to find $f(x|y)$, where $y = 1$. First, we find that

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Now

$$f(x|1) = \frac{f(x, 1)}{h(1)} = \left(\frac{7}{3}\right) f(x, 1), \quad x = 0, 1, 2.$$

Therefore,

$$f(0|1) = \left(\frac{7}{3}\right) f(0, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2}, \quad f(1|1) = \left(\frac{7}{3}\right) f(1, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2},$$

$$f(2|1) = \left(\frac{7}{3}\right) f(2, 1) = \left(\frac{7}{3}\right) (0) = 0,$$

and the conditional distribution of X , given that $Y = 1$, is

x	0	1	2
$f(x 1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

Finally,

$$P(X = 0 | Y = 1) = f(0|1) = \frac{1}{2}.$$

Therefore, if it is known that 1 of the 2 pen refills selected is red, we have a probability equal to $1/2$ that the other refill is not blue. ■

Example 3.19: The joint density for the random variables (X, Y) , where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- Find the marginal densities $g(x)$, $h(y)$, and the conditional density $f(y|x)$.
- Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Solution: (a) By definition,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy$$

$$= \frac{10}{3} xy^3 \Big|_{y=x}^{y=1} = \frac{10}{3} x(1 - x^3), \quad 0 < x < 1,$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = 5x^2 y^2 \Big|_{x=0}^{x=y} = 5y^4, \quad 0 < y < 1.$$

Now

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1 - x^3)} = \frac{3y^2}{1 - x^3}, \quad 0 < x < y < 1.$$

(b) Therefore,

$$P\left(Y > \frac{1}{2} \mid X = 0.25\right) = \int_{1/2}^1 f(y | x = 0.25) dy = \int_{1/2}^1 \frac{3y^2}{1 - 0.25^3} dy = \frac{8}{9}. \quad \text{■}$$

Example 3.20: Given the joint density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, \quad 0 < y < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find $g(x)$, $h(y)$, $f(x|y)$, and evaluate $P(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3})$.

Solution: By definition of the marginal density, for $0 < x < 2$,

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^1 \frac{x(1+3y^2)}{4} \, dy \\ &= \left(\frac{xy}{4} + \frac{xy^3}{4} \right) \Big|_{y=0}^{y=1} = \frac{x}{2}, \end{aligned}$$

and for $0 < y < 1$,

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^2 \frac{x(1+3y^2)}{4} \, dx \\ &= \left(\frac{x^2}{8} + \frac{3x^2y^2}{8} \right) \Big|_{x=0}^{x=2} = \frac{1+3y^2}{2}. \end{aligned}$$

Therefore, using the conditional density definition, for $0 < x < 2$,

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{x(1+3y^2)/4}{(1+3y^2)/2} = \frac{x}{2},$$

and

$$P\left(\frac{1}{4} < X < \frac{1}{2} \mid Y = \frac{1}{3}\right) = \int_{1/4}^{1/2} \frac{x}{2} \, dx = \frac{3}{64}.$$



Statistical Independence

If $f(x|y)$ does not depend on y , as is the case for Example 3.20, then $f(x|y) = g(x)$ and $f(x, y) = g(x)h(y)$. The proof follows by substituting

$$f(x, y) = f(x|y)h(y)$$

into the marginal distribution of X . That is,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_{-\infty}^{\infty} f(x|y)h(y) \, dy.$$

If $f(x|y)$ does not depend on y , we may write

$$g(x) = f(x|y) \int_{-\infty}^{\infty} h(y) \, dy.$$

Now

$$\int_{-\infty}^{\infty} h(y) \, dy = 1,$$

since $h(y)$ is the probability density function of Y . Therefore,

$$g(x) = f(x|y) \quad \text{and then} \quad f(x, y) = g(x)h(y).$$

It should make sense to the reader that if $f(x|y)$ does not depend on y , then of course the outcome of the random variable Y has no impact on the outcome of the random variable X . In other words, we say that X and Y are independent random variables. We now offer the following formal definition of statistical independence.

Definition 3.12:

Let X and Y be two random variables, discrete or continuous, with joint probability distribution $f(x, y)$ and marginal distributions $g(x)$ and $h(y)$, respectively. The random variables X and Y are said to be **statistically independent** if and only if

$$f(x, y) = g(x)h(y)$$

for all (x, y) within their range.

The continuous random variables of Example 3.20 are statistically independent, since the product of the two marginal distributions gives the joint density function. This is obviously not the case, however, for the continuous variables of Example 3.19. Checking for statistical independence of discrete random variables requires a more thorough investigation, since it is possible to have the product of the marginal distributions equal to the joint probability distribution for some but not all combinations of (x, y) . If you can find any point (x, y) for which $f(x, y)$ is defined such that $f(x, y) \neq g(x)h(y)$, the discrete variables X and Y are not statistically independent.

Example 3.21:

Show that the random variables of Example 3.14 are not statistically independent.

Proof: Let us consider the point $(0, 1)$. From Table 3.1 we find the three probabilities $f(0, 1)$, $g(0)$, and $h(1)$ to be

$$\begin{aligned} f(0, 1) &= \frac{3}{14}, \\ g(0) &= \sum_{y=0}^2 f(0, y) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14}, \\ h(1) &= \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}. \end{aligned}$$

Clearly,

$$f(0, 1) \neq g(0)h(1),$$

and therefore X and Y are not statistically independent. ▮

All the preceding definitions concerning two random variables can be generalized to the case of n random variables. Let $f(x_1, x_2, \dots, x_n)$ be the joint probability function of the random variables $X_1, X_2, \dots, X_n$. The marginal distribution of X_1 , for example, is

$$g(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for the discrete case, and

$$g(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_2 dx_3 \cdots dx_n$$

for the continuous case. We can now obtain **joint marginal distributions** such as $g(x_1, x_2)$, where

$$g(x_1, x_2) = \begin{cases} \sum_{x_3} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n) & \text{(discrete case),} \\ \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_3 dx_4 \cdots dx_n & \text{(continuous case).} \end{cases}$$

We could consider numerous conditional distributions. For example, the **joint conditional distribution** of X_1, X_2 , and X_3 , given that $X_4 = x_4, X_5 = x_5, \dots, X_n = x_n$, is written

$$f(x_1, x_2, x_3 \mid x_4, x_5, \dots, x_n) = \frac{f(x_1, x_2, \dots, x_n)}{g(x_4, x_5, \dots, x_n)},$$

where $g(x_4, x_5, \dots, x_n)$ is the joint marginal distribution of the random variables $X_4, X_5, \dots, X_n$.

A generalization of Definition 3.12 leads to the following definition for the mutual statistical independence of the variables $X_1, X_2, \dots, X_n$.

Definition 3.13: Let $X_1, X_2, \dots, X_n$ be n random variables, discrete or continuous, with joint probability distribution $f(x_1, x_2, \dots, x_n)$ and marginal distribution $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$, respectively. The random variables $X_1, X_2, \dots, X_n$ are said to be mutually **statistically independent** if and only if

$$f(x_1, x_2, \dots, x_n) = f_1(x_1)f_2(x_2) \cdots f_n(x_n)$$

for all $(x_1, x_2, \dots, x_n)$ within their range.

Example 3.22: Suppose that the shelf life, in years, of a certain perishable food product packaged in cardboard containers is a random variable whose probability density function is given by

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Let X_1, X_2 , and X_3 represent the shelf lives for three of these containers selected independently and find $P(X_1 < 2, 1 < X_2 < 3, X_3 > 2)$.

Solution: Since the containers were selected independently, we can assume that the random variables X_1, X_2 , and X_3 are statistically independent, having the joint probability density

$$f(x_1, x_2, x_3) = f(x_1)f(x_2)f(x_3) = e^{-x_1}e^{-x_2}e^{-x_3} = e^{-x_1-x_2-x_3},$$

for $x_1 > 0, x_2 > 0, x_3 > 0$, and $f(x_1, x_2, x_3) = 0$ elsewhere. Hence

$$\begin{aligned} P(X_1 < 2, 1 < X_2 < 3, X_3 > 2) &= \int_2^{\infty} \int_1^3 \int_0^2 e^{-x_1-x_2-x_3} dx_1 dx_2 dx_3 \\ &= (1 - e^{-2})(e^{-1} - e^{-3})e^{-2} = 0.0372. \end{aligned}$$

What Are Important Characteristics of Probability Distributions and Where Do They Come From?

This is an important point in the text to provide the reader with a transition into the next three chapters. We have given illustrations in both examples and exercises of practical scientific and engineering situations in which probability distributions and their properties are used to solve important problems. These probability distributions, either discrete or continuous, were introduced through phrases like “it is known that” or “suppose that” or even in some cases “historical evidence suggests that.” These are situations in which the nature of the distribution and even a good estimate of the probability structure can be determined through historical data, data from long-term studies, or even large amounts of planned data. The reader should remember the discussion of the use of histograms in Chapter 1 and from that recall how frequency distributions are estimated from the histograms. However, not all probability functions and probability density functions are derived from large amounts of historical data. There are a substantial number of situations in which the nature of the scientific scenario suggests a distribution type. Indeed, many of these are reflected in exercises in both Chapter 2 and this chapter. When independent repeated observations are binary in nature (e.g., defective or not, survive or not, allergic or not) with value 0 or 1, the distribution covering this situation is called the **binomial distribution** and the probability function is known and will be demonstrated in its generality in Chapter 5. Exercise 3.34 in Section 3.3 and Review Exercise 3.80 are examples, and there are others that the reader should recognize. The scenario of a continuous distribution in time to failure, as in Review Exercise 3.69 or Exercise 3.27 on page 93, often suggests a distribution type called the **exponential distribution**. These types of illustrations are merely two of many so-called standard distributions that are used extensively in real-world problems because the scientific scenario that gives rise to each of them is recognizable and occurs often in practice. Chapters 5 and 6 cover many of these types along with some underlying theory concerning their use.

A second part of this transition to material in future chapters deals with the notion of **population parameters** or **distributional parameters**. Recall in Chapter 1 we discussed the need to use data to provide information about these parameters. We went to some length in discussing the notions of a **mean** and **variance** and provided a vision for the concepts in the context of a population. Indeed, the population mean and variance are easily found from the probability function for the discrete case or probability density function for the continuous case. These parameters and their importance in the solution of many types of real-world problems will provide much of the material in Chapters 8 through 17.

Exercises

3.37 Determine the values of c so that the following functions represent joint probability distributions of the random variables X and Y :

(a) $f(x, y) = cxy$, for $x = 1, 2, 3$; $y = 1, 2, 3$;

(b) $f(x, y) = c|x - y|$, for $x = -2, 0, 2$; $y = -2, 3$.

3.38 If the joint probability distribution of X and Y is given by

$$f(x, y) = \frac{x + y}{30}, \quad \text{for } x = 0, 1, 2, 3; \ y = 0, 1, 2,$$

find